

SQL Self-Study Syllabus

Foundations of SQL for Data Analysis

Duration: 12 weeks | **Weekly:** ~2.5 hrs (Sat 11:15–12:45 + optional Sun 10:00–11:00)

Goal: Working proficiency with joins, aggregation, subqueries and data manipulation – finishing with a small portfolio project on real or sample data.

Resources

Tag	Resource	Use
Book	Sams Teach Yourself SQL in 10 Minutes (Ben Forta) – 22 lessons	Primary reading & concept reference
SQLBolt	SQLBolt interactive lessons (sqlbolt.com)	Hands-on practice in the browser
Video	One free video course (see picks below)	Reinforce each topic visually
Sandbox	SQLite + sample DB (e.g. Chinook) or your SIS read-access	Apply skills to realistic data

Video course picks (choose one, sample first)

- **TechTFQ** (YouTube) – topic-by-topic, clear logical explanations, includes a PostgreSQL setup guide. Best fit for this structure.
- **Alex the Analyst** (YouTube) – short ~10-min videos per topic, analyst-focused.
- **freeCodeCamp** (YouTube) – comprehensive long-form single-video courses; good as a jump-around reference.
- **Data with Baraa** (YouTube) – extensive long-form playlists, free version of paid content.

Tip: match one focused video to the topic you're reading that week rather than committing to a single 20+ hour marathon. Sample two channels in Week 1 and lock in whichever style clicks.

How each week works

Each week follows the same rhythm so it becomes automatic:

- **Read** the assigned book lesson(s) – short, ~10 min each.
- **Watch** a video segment on the same topic.
- **Practice** the matching SQLBolt lesson(s).
- **Apply** – write 2–3 of your own queries against the sample DB (or your SIS data).
- **Log** one "useful query" to your running personal cheat-sheet.

The optional Sunday hour is a buffer / catch-up slot – use it only if you fell behind or want extra practice. Don't treat it as mandatory.

Module 1 – Retrieving & Sorting Data (Weeks 1–2)

Week 1 – Getting oriented + basic retrieval

Book: Lesson 1 (Understanding SQL), Lesson 2 (Retrieving Data)

Video: Intro / what is SQL + SELECT basics; install SQLite or set up sample DB

SQLBolt: Lessons 1–2 (SELECT queries)

Apply: Set up your sandbox. Run SELECT with LIMIT on a few tables just to see the data.

★ Milestone: Environment working; can retrieve specific columns from a table.

Week 2 – Sorting & first filtering

Book: Lesson 3 (Sorting Retrieved Data), Lesson 4 (Filtering Data)

Video: ORDER BY + WHERE

SQLBolt: Lessons 3–4 (ORDER BY, WHERE filtering)

Apply: Write a query like "all records in category X, sorted by date."

★ Milestone: Comfortable with SELECT ... WHERE ... ORDER BY.

Module 2 – Filtering Deeper (Weeks 3–4)

Week 3 – Advanced filtering

Book: Lesson 5 (Advanced Data Filtering), Lesson 6 (Wildcard Filtering)

Video: AND/OR/IN/NOT + LIKE wildcards

SQLBolt: Lessons 5–6

Apply: Combine conditions – "status = A AND date > X, name starting with M."

★ Milestone: Can build multi-condition filters confidently.

Week 4 – Calculated fields

Book: Lesson 7 (Creating Calculated Fields)

Video: Concatenation, aliases, arithmetic in SELECT

SQLBolt: Review lessons 1–6; redo any that felt shaky

Apply: Create a derived column (full name from first + last, or a computed total).

★ Milestone: Comfortable shaping output, not just retrieving it. End of Module 2 – consolidate before functions.

Module 3 – Functions & Summarizing (Weeks 5–6)

Week 5 – Data manipulation functions

Book: Lesson 8 (Using Data Manipulation Functions)

Video: Text, numeric, and date functions

SQLBolt: Continue interactive review / any function-related

Apply: Use a date function to find records from "this month"; format a text field.

★ Milestone: Know where to look for the right function (and that dialects differ).

Week 6 – Summarizing data

Book: Lesson 9 (Summarizing Data)

Video: Aggregate functions – COUNT, SUM, AVG, MIN, MAX

SQLBolt: Aggregate functions lesson

Apply: "How many records per category", "average of column X."

★ Milestone: Can answer "how many / how much" questions. SQL starts feeling genuinely useful here.

Module 4 – Grouping & Subqueries (Weeks 7–8)

Week 7 – Grouping data

Book: Lesson 10 (Grouping Data)

Video: GROUP BY + HAVING

SQLBolt: Aggregate functions w/ GROUP BY lesson

Apply: "Count of students per grade", then filter groups with HAVING.

★ Milestone: Comfortable with GROUP BY ... HAVING.

Week 8 – Subqueries

Book: Lesson 11 (Working with Subqueries)

Video: Subqueries / nested SELECTs

SQLBolt: Review; practice nesting a query inside a WHERE

Apply: "Find records where value is above the overall average" (subquery in WHERE).

★ Milestone: Can use a subquery to answer a two-step question.

Module 5 – Joins (Weeks 9–10) ★ the big one

Joins are the conceptual centrepiece. Don't rush – if it takes both weekend slots, that's expected and fine.

Week 9 – Inner joins

Book: Lesson 12 (Joining Tables)

Video: INNER JOIN explained (watch 2 different explanations if it doesn't click)

SQLBolt: Lessons on multi-table queries with JOINS

Apply: Join two related tables from your data (e.g. students + enrollments).

★ Milestone: Understand why joins matter and can write a basic INNER JOIN.

Week 10 – Advanced joins & combining queries

Book: Lesson 13 (Creating Advanced Joins), Lesson 14 (Combining Queries)

Video: LEFT/OUTER joins + UNION

SQLBolt: OUTER JOIN lessons

Apply: "Show all students who have NO emergency contact" (LEFT JOIN + NULL check).

★ Milestone: Can pick the right join type for the question. End of Module 5 – you're now genuinely useful at work.

Module 6 – Modifying Data & Wrap-up (Weeks 11–12)

Week 11 – Inserting, updating, deleting + tables

Book: Lesson 15 (Inserting Data), Lesson 16 (Updating & Deleting Data), Lesson 17 (Creating & Manipulating Tables)

Video: INSERT / UPDATE / DELETE + CREATE TABLE

SQLBolt: Lessons on INSERT, UPDATE, DELETE, and creating tables

Apply: In your sandbox only, create a small table, insert rows, update and delete some.

★ Milestone: Understand the full CRUD cycle (and the danger of UPDATE/DELETE without WHERE).

△ Never run UPDATE/DELETE against live work data while learning.

Week 12 – Views + capstone project

Book: Lesson 18 (Using Views); skim Lessons 19–22 for awareness – not mastery

Video: Views (and a skim of what stored procedures are)

Capstone: Pick a question that matters to your team (or uses the sample DB); write a multi-table query (join + aggregate + filter) to answer it; save it, document it in plain English, add it to your cheat-sheet.

★ **Milestone:** A finished query/mini-analysis you can talk about in an interview or show your team.

After Week 12 – where next

- **Window functions** (ROW_NUMBER, RANK, running totals) – the natural "intermediate" next step; high-value for analyst work.
- **Python + pandas** – the planned follow-on; pandas will feel familiar after SQL.
- **Keep the cheat-sheet alive** – your running list of useful queries is the most valuable artifact you'll build here.

Progress tracker

Wk	Topic	Book	Bolt	Video	Applied	Done
1	Intro + retrieval	☐	☐	☐	☐	☐
2	Sorting + filtering	☐	☐	☐	☐	☐
3	Advanced filtering	☐	☐	☐	☐	☐
4	Calculated fields	☐	☐	☐	☐	☐
5	Functions	☐	☐	☐	☐	☐
6	Summarizing	☐	☐	☐	☐	☐
7	Grouping	☐	☐	☐	☐	☐
8	Subqueries	☐	☐	☐	☐	☐
9	Inner joins	☐	☐	☐	☐	☐
10	Advanced joins	☐	☐	☐	☐	☐
11	Insert/update/delete	☐	☐	☐	☐	☐
12	Views + capstone	☐	☐	☐	☐	☐